

School of Computer Science and Artificial Intelligence

Lab Assignment # 6.5

Program	: B. Tech (CSE)
Specialization	:
Course Title	: AI Assisted coding
Course Code	:
Semester	: II
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Task 1: Use an AI tool to generate eligibility logic.

Prompt : Generate Python code to check voting eligibility based on age and citizenship.

Code :

```
▶ age = int(input("Enter age: "))
  citizen = input("Citizen (yes/no): ")
  if age >= 18 and citizen == "yes":
      print("Eligible to vote")
  else:
      print("Not eligible")
```

Output:

```
... Enter age: 18
    Citizen (yes/no): yes
    Eligible to vote
```

Explanation :

- The program takes age as integer input from the user.
- It takes citizenship status as string input (yes or no).
- The if statement checks two conditions together.
- First condition verifies whether age is 18 or above.
- Second condition checks if the user is a citizen.

- The and operator ensures both conditions must be true.
- If both are true, voting eligibility is confirmed.
- The program prints "Eligible to vote".
- If any condition fails, control goes to else.
- It prints "Not eligible" when criteria are not met.

Task 2 : Count Vowels and Consonants (Loop + String)

Code :

```
s = input("Enter string: ").lower()
v = c = 0
for ch in s:
    if ch.isalpha():
        if ch in "aeiou":
            v += 1
        else:
            c += 1
print("Vowels:", v, "Consonants:", c)
```

Output :

```
... Enter string: kaizen
Vowels: 3 Consonants: 3
```

Explanation :

- The program takes a string input from the user.
- The string is converted to lowercase for easy comparison.
- Two variables v and c are initialized to count vowels and consonants.
- A for loop iterates through each character in the string.
- isalpha() checks whether the character is a letter.
- Non-alphabet characters like spaces and numbers are ignored.
- The if condition checks if the character is a vowel.
- Vowels are counted using variable v.
- Remaining letters are counted as consonants using c.
- The final count of vowels and consonants is displayed.

Task 3: Library Management System (Class + Loop + Condition)

Prompt: Generate a Python program for a library management system using classes, loops, and conditional statements.

Code :

```
class Library:
    def __init__(self):
        self.books = []
    def add_book(self, b):
        self.books.append(b)
    def show_books(self):
        for b in self.books:
            print(b)

lib = Library()
while True:
    ch = input("1.Add 2.Show 3.Exit: ")
    if ch == "1": lib.add_book(input("Book name: "))
    elif ch == "2": lib.show_books()
    else: break
```

Output :

```
... 1.Add 2.Show 3.Exit: 1
    Book name: kaizen
```

Explanation :

- The program defines a class named Library.
- The constructor initializes an empty list to store books.
- add_book() method adds a new book to the list.
- show_books() method displays all stored books.
- An object lib is created from the Library class.
- A while True loop keeps the program running continuously.
- The user is shown a menu with add, show, and exit options.
- If choice is 1, a book name is added to the library.
- If choice is 2, all books are displayed.
- Any other choice exits the program.

Task 4: Attendance Management System (Class + Loop)

Prompt: Generate a Python class to mark and display student attendance using loops.

Code:

```
class Attendance:
    def __init__(self):
        self.data = {}
    def mark(self, name):
        self.data[name] = "Present"
    def display(self):
        for k, v in self.data.items():
            print(k, ":", v)

a = Attendance()
a.mark("Sai")
a.mark("Akash")
a.display()
```

Output :

```
... Sai : Present
    Akash : Present
```

Explanation :

- The program defines a class named Attendance.
- The constructor initializes an empty dictionary to store attendance data.
- The dictionary stores student names as keys.
- Attendance status is stored as values in the dictionary.
- The mark() method marks a student as "Present".
- The display() method prints all attendance records.
- A for loop is used to access dictionary items.
- An object a is created from the Attendance class.
- Attendance is marked for students Sai and Ravi.
- The final attendance list is displayed on the screen.

Task 5: ATM Menu Simulation (Loop + Conditionals)

Prompt: Generate a Python program using loops and conditionals to simulate an ATM menu.
Code:

```
bal = 1000
while True:
    ch = input("1.Deposit 2.Withdraw 3.Balance 4.Exit: ")
    if ch == "1": bal += int(input("Amount: "))
    elif ch == "2": bal -= int(input("Amount: "))
    elif ch == "3": print("Balance:", bal)
    else: break
```

Output :

```
... 1.Deposit 2.Withdraw 3.Balance 4.Exit: 1
Amount: 5500
1.Deposit 2.Withdraw 3.Balance 4.Exit: 2
Amount: 2000
1.Deposit 2.Withdraw 3.Balance 4.Exit: 3
Balance: 4500
1.Deposit 2.Withdraw 3.Balance 4.Exit: 4
```

Explanation :

- The program initializes the account balance with 1000.
- A while True loop keeps the ATM menu running.
- The user is shown options for deposit, withdraw, balance, and exit.
- User input is stored in the variable ch.
- If the choice is 1, the entered amount is added to the balance.
- If the choice is 2, the entered amount is subtracted from the balance.
- If the choice is 3, the current balance is displayed.
- Conditional statements control the menu flow.
- The loop repeats after each operation.
- The program stops when the user selects exit.